Homework 19.10.26 – Exercises 20, 24, 25

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Exercise 20

Given that f has L-Lipschitz gradient, i.e.,

$$\|\nabla f(x) - \nabla f(y)\| \le L\,\|x-y\| \qquad \forall x, y \in \mathbb{R}^d.$$

Show that

$$-\frac{L}{2}\,\|x-y\|^2 \le D_f(x,y) \le \frac{L}{2}\,\|x-y\|^2 \qquad \forall x, y \in \mathbb{R}^d.$$

Proof

It is enough to show that $|D_f(x,y)| \le \frac{L}{2}\,\|x-y\|^2$. Let $z := y + t(x-y)$, so that $x - y = \frac{z-y}{t}$. Then

$$\begin{aligned}
|D_f(x,y)| &= \big|f(x) - f(y) - \langle \nabla f(y), x-y\rangle\big| \\
&= \left|\int_0^1 \big\langle \nabla f\big(y + t(x-y)\big), x-y\big\rangle\, dt - \langle \nabla f(y), x-y\rangle\right| \\
&= \left|\int_0^1 \langle \nabla f(z) - \nabla f(y), z-y\rangle \cdot \frac{1}{t}\, dt\right| \le \int_0^1 \frac{1}{t}\,\big|\langle \nabla f(z) - \nabla f(y), z-y\rangle\big|\, dt.
\end{aligned}$$

By Cauchy–Schwarz and then by the L-Lipschitz gradient assumption,

$$\begin{aligned}
|D_f(x,y)| &\le \int_0^1 \frac{1}{t}\,\|z-y\|\,\|\nabla f(z) - \nabla f(y)\|\, dt \le \int_0^1 \frac{L}{t}\,\|z-y\|^2\, dt = \int_0^1 \frac{L}{t}\,\langle z-y, z-y\rangle\, dt \\
&= \int_0^1 L\,\langle x-y, x-y\rangle\, t\, dt = \frac{L}{2}\,\|x-y\|^2,
\end{aligned}$$

where we used $z - y = t(x-y)$. $\Rightarrow -\frac{L}{2}\,\|x-y\|^2 \le D_f(x,y) \le \frac{L}{2}\,\|x-y\|^2$.

Exercise 24

For all $a, b \in \mathbb{R}^d$ and $t > 0$ the following inequalities hold:

$$\begin{aligned}
&1) \quad \langle a, b\rangle \le \frac{\|a\|^2}{2t} + \frac{t\,\|b\|^2}{2}, \\
&2) \quad \|a+b\|^2 \le 2\,\|a\|^2 + 2\,\|b\|^2, \\
&3) \quad \frac{1}{2}\,\|a\|^2 - \|b\|^2 \le \|a+b\|^2.
\end{aligned}$$

Proof of 1)

Let $a' = \frac{a}{\sqrt{t}}$, $b' = b\sqrt{t}$, i.e., $a = a'\sqrt{t}$, $b = \frac{b'}{\sqrt{t}}$. Then

$$\langle a, b\rangle = \langle a', b'\rangle, \qquad \frac{\|a\|^2}{2t} = \frac{\|a'\|^2}{2}, \qquad \frac{t\,\|b\|^2}{2} = \frac{\|b'\|^2}{2}.$$

As a result, we only have to prove

$$2\langle a', b'\rangle \le \|a'\|^2 + \|b'\|^2,$$

which is equivalent to $\|a' - b'\|^2 \ge 0$, so it is true.

Proof of 2)

Using $\|a - b\|^2 \ge 0$ we get

$$\|a\|^2 + \|b\|^2 - 2\langle a, b\rangle \ge 0.$$

Adding $\|a\|^2 + \|b\|^2$ to both sides:

$$2\|a\|^2 + 2\|b\|^2 - 2\langle a, b\rangle \ge \|a\|^2 + \|b\|^2.$$

Adding $2\langle a, b\rangle$ and regrouping:

$$2\|a\|^2 + 2\|b\|^2 \ge \|a\|^2 + 2\langle a, b\rangle + \|b\|^2 = \|a + b\|^2.$$

Proof of 3)

3) is the same as

$$\|a\|^2 \le 2\|a + b\|^2 + 2\|b\|^2.$$

Setting $a := a' + b'$, $b := -b'$ (so that $a + b = a'$) we get

$$\|a' + b'\|^2 \le 2\|a'\|^2 + 2\|b'\|^2,$$

which is the same as 2).

Exercise 25

For a random vector $X \in \mathbb{R}^d$ (with finite second moment) and any $c \in \mathbb{R}^d$,

$$\mathbb{E}\big[\|X - \mathbb{E}[X]\|^2\big] = \mathbb{E}\big[\|X - c\|^2\big] - \|\mathbb{E}[X] - c\|^2.$$

Proof

$$\begin{aligned}\mathbb{E}\big[\|X - \mathbb{E}[X]\|^2\big] &= \mathbb{E}\big[\|(X - c) - (\mathbb{E}[X] - c)\|^2\big] \\ &= \mathbb{E}\big[\|X - c\|^2\big] + \mathbb{E}\big[\|\mathbb{E}[X] - c\|^2\big] - 2\,\mathbb{E}\big[\langle X - c, \mathbb{E}[X] - c\rangle\big].\end{aligned}$$

On the other hand, $\|\mathbb{E}[X] - c\|^2$ is a number, so its expectation is the same number:

$$\mathbb{E}\big[\|\mathbb{E}[X] - c\|^2\big] = \|\mathbb{E}[X] - c\|^2.$$

So we get

$$\mathbb{E}\big[\|X - \mathbb{E}[X]\|^2\big] = \mathbb{E}\big[\|X - c\|^2\big] - \|\mathbb{E}[X] - c\|^2 - 2\Big(\mathbb{E}\big[\langle X - c, \mathbb{E}[X] - c\rangle\big] - \|\mathbb{E}[X] - c\|^2\Big).$$

Also

$$\begin{aligned}\mathbb{E}\big[\langle X-c, \mathbb{E}[X]-c\rangle\big] &= \mathbb{E}\big[\langle X, \mathbb{E}[X]\rangle\big] - \langle c, \mathbb{E}[X]\rangle - \langle c, \mathbb{E}[X]\rangle + \|c\|^2 \\ &= \|\mathbb{E}[X]\|^2 - 2\,\langle c, \mathbb{E}[X]\rangle + \|c\|^2 = \|\mathbb{E}[X]-c\|^2\,,\end{aligned}$$

which means

$$-2\Big(\mathbb{E}\big[\langle X-c, \mathbb{E}[X]-c\rangle\big] - \|\mathbb{E}[X]-c\|^2\Big) = 0,$$

which proves the identity.